

Deep Learning and Its Application

Week 12: Practicals

1. Write a code in Python to create a two folder handout. The following code fragment is used to generate a sequence of integers from 0 to 19,

```
2. from random import seed
   from random import choice
   import numpy as np
   from sklearn.model_selection import KFold
   # seed random number generator
   seed(1)
   # prepare a sequence
   sequence = [i for i in range(20)]
```

- a) Use the 'KFold' function in Scikit-Learn to create two folders for training and test
- b) Print out the training and test data in these folders

Answer:

```
from random import seed
from random import choice
import numpy as np
from sklearn.model_selection import KFold
# seed random number generator
seed(1)
# prepare a sequence
sequence = [i for i in range(20)]
print(sequence)

X = np.array(sequence)
kf = KFold(n_splits=2, random_state=None, shuffle=False)
#kf = KFold(n_splits=2, random_state=True, shuffle=True)

for train, test in kf.split(X):
    print('train: %s, test: %s' % (X[train], X[test]))
```

2. The following code is designed for the MNIST handwritten digit classification, please refer to Lecture 8

```
from numpy import mean
from numpy import std
from sklearn.model_selection import KFold
from tensorflow.keras.datasets import mnist
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D
from tensorflow.keras.layers import MaxPooling2D
from tensorflow.keras.layers import Dense
from tensorflow.keras.layers import Flatten
from tensorflow.keras.optimizers import SGD

# load train and test dataset
def load_dataset():
    # load dataset
    (trainX, trainY), (testX, testY) = mnist.load_data()
    # reshape dataset to have a single channel
```

```

trainX = trainX.reshape((trainX.shape[0], 28, 28, 1))
testX = testX.reshape((testX.shape[0], 28, 28, 1))
# one hot encode target values
trainY = to_categorical(trainY)
testY = to_categorical(testY)
return trainX, trainY, testX, testY

# scale pixels
def prep_pixels(train, test):
    # convert from integers to floats
    train_norm = train.astype('float32')
    test_norm = test.astype('float32')
    # normalize to range 0-1
    train_norm = train_norm / 255.0
    test_norm = test_norm / 255.0
    # return normalized images
    return train_norm, test_norm

# define cnn model
def define_model():
    model = Sequential()
    model.add(Conv2D(32, (3, 3), activation='relu', kernel_initializer='he_uniform',
input_shape=(28, 28, 1)))
    model.add(MaxPooling2D((2, 2)))
    model.add(Flatten())
    model.add(Dense(100, activation='relu', kernel_initializer='he_uniform'))
    model.add(Dense(10, activation='softmax'))
    # compile model
    opt = SGD(lr=0.01, momentum=0.9)
    model.compile(optimizer=opt, loss='categorical_crossentropy',
metrics=['accuracy'])
    return model

# evaluate a model using k-fold cross-validation
def evaluate_model(dataX, dataY, n_folds=5):
    scores, histories = list(), list()
    # prepare cross validation
    kfold = KFold(n_folds, shuffle=True, random_state=1)
    # enumerate splits
    for train_ix, test_ix in kfold.split(dataX):
        .
        .
        .
        # select rows for train and test
        trainX, trainY, testX, testY = dataX[train_ix], dataY[train_ix],
dataX[test_ix], dataY[test_ix]
        # fit model
        history = model.fit(trainX, trainY, epochs=10, batch_size=32,
validation_data=(testX, testY), verbose=0)
        # evaluate model
        _, acc = model.evaluate(testX, testY, verbose=0)
        print('> %.3f' % (acc * 100.0))
        # stores scores
        scores.append(acc)
        histories.append(history)
    return scores, histories

```

- Complete the “evaluate_model” function with one statement of invoking “define_model”
- Write a small code to call the functions of “load_dataset”, “prep_pixels” and “evaluate_model”
- Save the classification model into your local drive

- d. Download “sample_image” file from the Week 12 folder in BBL into your local drive
- e. Below is a code fragment for loading the image of “sample_image” from your local drive, write a small code to load the saved classification model in (c) above and predict “sample_image”

```
from tensorflow.keras.preprocessing.image import load_img
from tensorflow.keras.preprocessing.image import img_to_array
from tensorflow.keras.models import load_model

# load and prepare the image
def load_image(filename):
    # load the image
    img = load_img(filename, grayscale=True, target_size=(28, 28))
    # convert to array
    img = img_to_array(img)
    # reshape into a single sample with 1 channel
    img = img.reshape(1, 28, 28, 1)
    # prepare pixel data
    img = img.astype('float32')
    img = img / 255.0
    return img
```

Answer:

a)

```
model = define_model()
# select rows for train and test
```

b)

```
trainX, trainY, testX, testY = load_dataset()
# prepare pixel data
trainX, testX = prep_pixels(trainX, testX)
# evaluate model
scores, histories = evaluate_model(trainX, trainY)
```

c)

```
trainX, trainY, testX, testY = load_dataset()
# prepare pixel data
trainX, testX = prep_pixels(trainX, testX)
# define model
model = define_model()
# fit model
model.fit(trainX, trainY, epochs=10, batch_size=32, verbose=0)
# save model
model.save('final_model.h5')
```

e)

```
# load an image and predict the class
def run_example():
```

```

# load the image
img = load_image('sample_image.png')
# load model
model = load_model('final_model.h5')
# predict the class
digit = model.predict_classes(img)
print(digit[0])

# entry point, run the example
run_example()

```

3. Please refer to the Model Improvement in Lecture 12, enhance the convolution layer in the “define-model” function in Q2 above

Answer:

```

def define_model():
    model = Sequential()
    model.add(Conv2D(32, (3, 3), activation='relu', kernel_initializer='he_uniform',
input_shape=(28, 28, 1)))
    model.add(MaxPooling2D((2, 2)))
    model.add(Conv2D(64, (3, 3), activation='relu',
kernel_initializer='he_uniform'))
    model.add(Conv2D(64, (3, 3), activation='relu',
kernel_initializer='he_uniform'))
    model.add(MaxPooling2D((2, 2)))
    model.add(Flatten())
    model.add(Dense(100, activation='relu', kernel_initializer='he_uniform'))
    model.add(Dense(10, activation='softmax'))
    # compile model
    opt = SGD(lr=0.01, momentum=0.9)
    model.compile(optimizer=opt, loss='categorical_crossentropy',
metrics=['accuracy'])
    return model

```